Elita Lestari Setianto (201344008)	**Rangkuman Minggu 2**	Senin, 5 September 2022
		16TNK5043 (Aplikasi Mikrokontroler)
3NK		Ferry Satria, BSEE.,M.T. Rahmawati Hasanah,S.ST.,M.T.

SOAL (PENJUMLAHAN DESIMAL)

1. Buat program untuk menjumlahkan bilangan desimal berikut
    a. 78 + 95 disimpan pada register R3, R2
       R2 ratusan, R3 ribuan
    b. 8937 + 7899 = (r7, r6, r5)

Kesulitan

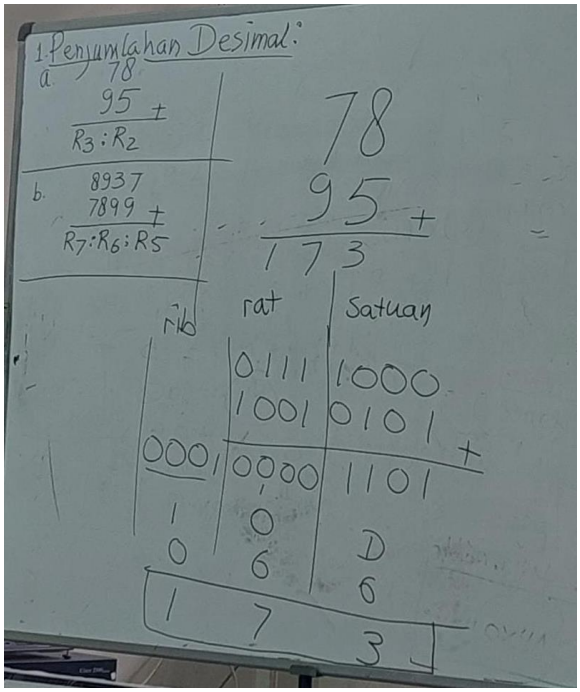
- Atmega8535 itu pada instruction set nya tidak disediakan perintah faktor koreksi supaya hasilnya desimal, karena yang dijumlahkan itu bilangan desimal dan sedangkan yang menghitung prosessor dan prosesor itu digital maka harus dihitung ke dalam bilangan biner

Solusi

Yaitu dengan menggunakan BCD Adder dengan menggunakan aturan subrootin

Kapan ditambah 6 ?

- Ketika nilai lebih dari 9
- Kalau dia memberikan carry ke yang kiri (carry lebih dr 15)



DAA (Desimal Adjust After

Subrutin bisa digunakan untuk faktor koreksi ketika bilangan BCD Adder itu 2 digit

Program Subrutin

```
; subrutin unt koreksi hasil penjumlahan mjd. BCD.2 dgt.
; yang dilaksanakan mll. r19 sbg. acc.:
daa:    push r16;amankan isi reg. ke stack
                push r17
                push r21
                push r24
                in r16,sreg; amankan sreg
                in r17,sreg
                ldi r24,0 ; r24 hanya sbg status dr carry flag
                adc r24,r24; status carry flag sebelum koreksi
; ambil, amankan kembali sreg, uji half carry
                out sreg,r17
                brhc lup4; lompat ke lup4 jika H=0
                rcall fk_ln; satuan + 06 jika H=1
        ;       rjmp lup5
lup4:   mov r21,r19
```

```
                andi r21,$0f; ambil dgt.satuan
                cpi r21,$0a ; > 9
                brcs lup5; lompat ke lup5 jika < 9
                rcall fk_ln; satuan + 06 jika > 9
lup5:   out sreg,r17; ambil, amankan kembali sreg, uji carry
                brcc lup6; lompat ke lup6 jika C=0
                rcall fk_hn; puluhan + 60 jika C=1
        ;       rjmp lup7
lup6:   mov r21,r19
                andi r21,0xf0; ambil dgt.puluhan
                cpi r21,0xa0
                brcs lup7; lompat ke lup7 jika pul < a
                rcall fk_hn; pul + 60 jika > 9
lup7:   out sreg,r16; mengatur carry flag sblm keluar:
                clc; C = 0
                in r16,sreg ;amankan sreg stlh c=0
                cpi r24,0 ;menguji status
                breq lup8
                out sreg,r16;ambil sreg, set c
                sec; C = 1
                in r16,sreg; amankan sreg stlh c=1
lup8:
                out sreg,r16;status akhir
                pop r24;kembalikan isi reg. dr stack
                pop r21
                pop r17
                pop r16
                ret
fk_ln:  ldi r21,06; koreksi dgt.satuan
                out sreg,r16
                add r19,r21
                in r16,sreg
                brcc tuh
```

```
                ldi r21,0x01
                add r17,r21
tuh:    out sreg,r16
                adc r24,r24
                ret
fk_hn:  ldi r21,$60; koreksi dgt.puluhan
                out sreg,r16
                add r19,r21
                in r16,sreg
                adc r24,r24; amankan carry pd r24
                ret ; akhir dari prog. subrutin.
```

1. 78 + 95

A. Kode Program

```
.include "m8535def.inc"
.org 0x00
rjmp main
        ; PAKAI PROJECT "ROOT"
main :
ldi r20,low(ramend)
out spl,r20
ldi r20,high(ramend)
out sph,r20
ldi r20,0x78
mov r2,r20
ldi r20,0x95
mov r7,r20
mov r19,r2
add r19,r7
rcall daa ;(relative call-- manggil subrutin sebelummnya yg sudah disiapkan)
mov r2,r19
mov r19,r3
```

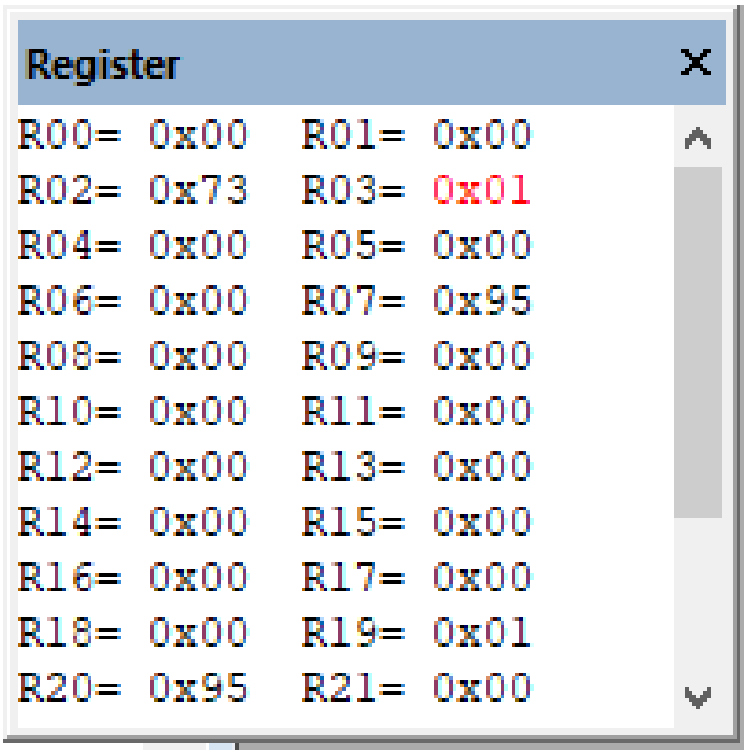
```
adc r19,r1 ;(tidak perlu dikoreksi karna nilainya tidak lebih dr 9)
mov r3,r19
akhir: rjmp akhir



; subrutin unt koreksi hasil penjumlahan mjd. BCD.2 dgt.
; yang dilaksanakan mll. r19 sbg. acc.:
daa:    push r16;amankan isi reg. ke stack
                push r17
                push r21
                push r24
                in r16,sreg; amankan sreg
                in r17,sreg
                ldi r24,0 ; r24 hanya sbg status dr carry flag
                adc r24,r24; status carry flag sebelum koreksi
; ambil, amankan kembali sreg, uji half carry
                out sreg,r17
                brhc lup4; lompat ke lup4 jika H=0
                rcall fk_ln; satuan + 06 jika H=1
        ;       rjmp lup5
lup4:   mov r21,r19
                andi r21,$0f; ambil dgt.satuan
                cpi r21,$0a ; > 9
                brcs lup5; lompat ke lup5 jika < 9
                rcall fk_ln; satuan + 06 jika > 9
lup5:   out sreg,r17; ambil, amankan kembali sreg, uji carry
                brcc lup6; lompat ke lup6 jika C=0
                rcall fk_hn; puluhan + 60 jika C=1
        ;       rjmp lup7
lup6:   mov r21,r19
                andi r21,0xf0; ambil dgt.puluhan
                cpi r21,0xa0
```

```
                brcs lup7; lompat ke lup7 jika pul < a
                rcall fk_hn; pul + 60 jika > 9
lup7:   out sreg,r16; mengatur carry flag sblm keluar:
                clc; C = 0
                in r16,sreg ;amankan sreg stlh c=0
                cpi r24,0 ;menguji status
                breq lup8
                out sreg,r16;ambil sreg, set c
                sec; C = 1
                in r16,sreg; amankan sreg stlh c=1
lup8:
                out sreg,r16;status akhir
                pop r24;kembalikan isi reg. dr stack
                pop r21
                pop r17
                pop r16
                ret
fk_ln:  ldi r21,06; koreksi dgt.satuan
                out sreg,r16
                add r19,r21
                in r16,sreg
                brcc tuh
                ldi r21,0x01
                add r17,r21
tuh:    out sreg,r16
                adc r24,r24
                ret
fk_hn:  ldi r21,$60; koreksi dgt.puluhan
                out sreg,r16
                add r19,r21
                in r16,sreg
                adc r24,r24; amankan carry pd r24
                ret ; akhir dari prog. subrutin.
```

B. Hasil Program



b.8937 + 7899 = (r7, r6, r5)

```
.include "m8535def.inc"
.org 0x00
rjmp main
        ; PAKAI PROJECT "ROOT"
main :
ldi r16,low(ramend)
out spl,r16                ; spl = stack pointer low
ldi r16,high(ramend)
out sph,r16

ldi r20,0x37
ldi r21,0x89    ; data pertama = 8937

ldi r22,0x99
ldi r23,0x78    ; data kedua = 7849
```

```
mov r19,r20
add r19,r22
rcall daa
mov r5,r19

mov r19,r21
adc r19,r23
rcall daa
mov r6,r19

adc r7,r1

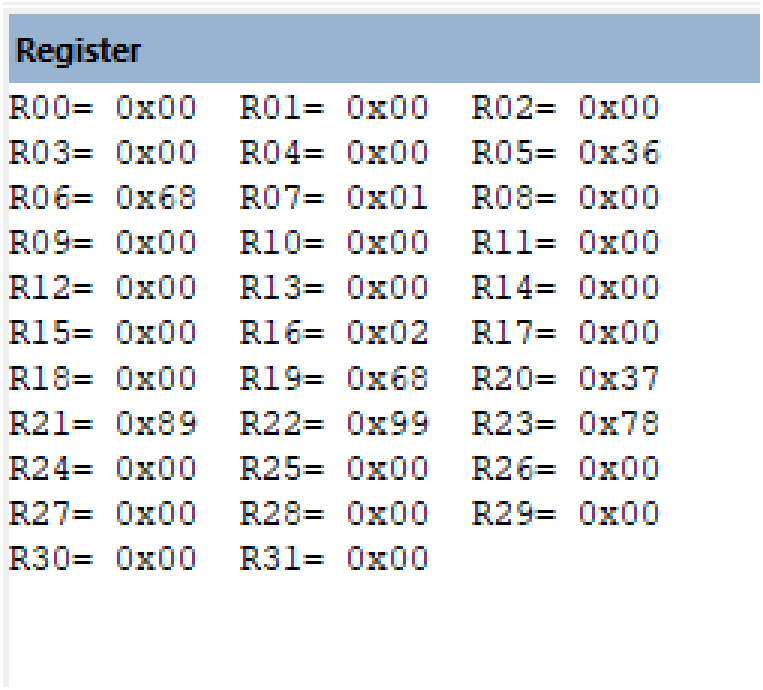
akhir: rjmp akhir

; subrutin unt koreksi hasil penjumlahan mjd. BCD.2 dgt.
; yang dilaksanakan mll. r19 sbg. acc.:
daa:    push r16;amankan isi reg. ke stack
                push r17
                push r21
                push r24
                in r16,sreg; amankan sreg
                in r17,sreg
                ldi r24,0 ; r24 hanya sbg status dr carry flag
                adc r24,r24; status carry flag sebelum koreksi
; ambil, amankan kembali sreg, uji half carry
                out sreg,r17
                brhc lup4; lompat ke lup4 jika H=0
                rcall fk_ln; satuan + 06 jika H=1
        ;       rjmp lup5
lup4:   mov r21,r19
                andi r21,$0f; ambil dgt.satuan
                cpi r21,$0a ; > 9
                brcs lup5; lompat ke lup5 jika < 9
```

```
                rcall fk_ln; satuan + 06 jika > 9
lup5:   out sreg,r17; ambil, amankan kembali sreg, uji carry
                brcc lup6; lompat ke lup6 jika C=0
                rcall fk_hn; puluhan + 60 jika C=1
        ;       rjmp lup7
lup6:   mov r21,r19
                andi r21,0xf0; ambil dgt.puluhan
                cpi r21,0xa0
                brcs lup7; lompat ke lup7 jika pul < a
                rcall fk_hn; pul + 60 jika > 9
lup7:   out sreg,r16; mengatur carry flag sblm keluar:
                clc; C = 0
                in r16,sreg ;amankan sreg stlh c=0
                cpi r24,0 ;menguji status
                breq lup8
                out sreg,r16;ambil sreg, set c
                sec; C = 1
                in r16,sreg; amankan sreg stlh c=1
lup8:   out sreg,r16;status akhir
                pop r24;kembalikan isi reg. dr stack
                pop r21
                pop r17
                pop r16
                ret
fk_ln:  ldi r21,06; koreksi dgt.satuan
                out sreg,r16
                add r19,r21
                in r16,sreg
                brcc tuh
                ldi r21,0x01
                add r17,r21
tuh:    out sreg,r16
                adc r24,r24
```

```
                ret
fk_hn:  ldi r21,$60; koreksi dgt.puluhan
                out sreg,r16
                add r19,r21
                in r16,sreg
                adc r24,r24; amankan carry pd r24
                ret ; akhir dari prog. subrutin.
```

Hasil program



2. Buatlah soal no 1 dengan menyimpan hasil di data area.
    - Harus memahami pointer

a. Kode program

```
.include "m8535def.inc"
.org 0x00
rjmp main
        ; PAKAI PROJECT "ROOT"
main :
ldi r16,low(ramend)
out spl,r16                ; spl = stack pointer low
```

```
ldi r16,high(ramend)
out sph,r16



ldi r28,0x60
ld r16,y+
ld r17,y

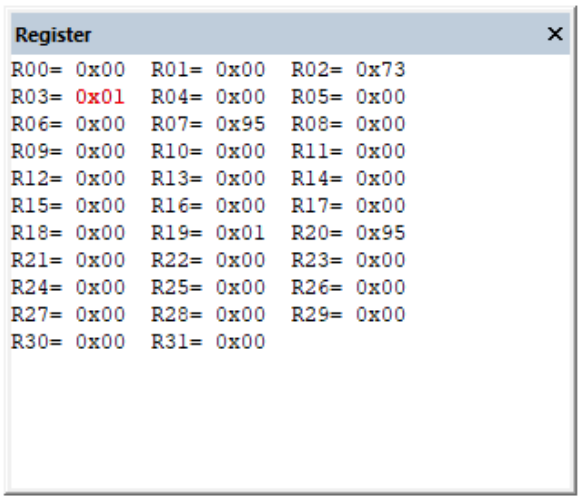
mov r2,r16
mov r7,r17
add r2,r7
mov r19,r2
rcall daa
mov r2,r19
mov r19,r3
adc r19,r1
mov r3,r19

akhir: rjmp akhir

; subrutin unt koreksi hasil penjumlahan mjd. BCD.2 dgt.
; yang dilaksanakan mll. r19 sbg. acc.:
daa:    push r16;amankan isi reg. ke stack
                push r17
                push r21
                push r24
                in r16,sreg; amankan sreg
                in r17,sreg
                ldi r24,0 ; r24 hanya sbg status dr carry flag
                adc r24,r24; status carry flag sebelum koreksi
; ambil, amankan kembali sreg, uji half carry
                out sreg,r17
                brhc lup4; lompat ke lup4 jika H=0
```

```
                rcall fk_ln; satuan + 06 jika H=1
        ;       rjmp lup5
lup4:   mov r21,r19
                andi r21,$0f; ambil dgt.satuan
                cpi r21,$0a ; > 9
                brcs lup5; lompat ke lup5 jika < 9
                rcall fk_ln; satuan + 06 jika > 9
lup5:   out sreg,r17; ambil, amankan kembali sreg, uji carry
                brcc lup6; lompat ke lup6 jika C=0
                rcall fk_hn; puluhan + 60 jika C=1
        ;       rjmp lup7
lup6:   mov r21,r19
                andi r21,0xf0; ambil dgt.puluhan
                cpi r21,0xa0
                brcs lup7; lompat ke lup7 jika pul < a
                rcall fk_hn; pul + 60 jika > 9
lup7:   out sreg,r16; mengatur carry flag sblm keluar:
                clc; C = 0
                in r16,sreg ;amankan sreg stlh c=0
                cpi r24,0 ;menguji status
                breq lup8
                out sreg,r16;ambil sreg, set c
                sec; C = 1
                in r16,sreg; amankan sreg stlh c=1
lup8:   out sreg,r16;status akhir
                pop r24;kembalikan isi reg. dr stack
                pop r21
                pop r17
                pop r16
                ret
fk_ln:  ldi r21,06; koreksi dgt.satuan
                out sreg,r16
                add r19,r21
```

```
                in r16,sreg
                brcc tuh
                ldi r21,0x01
                add r17,r21
tuh:    out sreg,r16
                adc r24,r24
                ret
fk_hn:  ldi r21,$60; koreksi dgt.puluhan
                out sreg,r16
                add r19,r21
                in r16,sreg
                adc r24,r24; amankan carry pd r24
                ret ; akhir dari prog. subrutin.
```



3. Buatlah soal no 1 dengan menyimpan hasil di program area.
    - Harus memahami pointer

b. Kode program

```
.include "m8535def.inc"
.org 0x00
rjmp main
        ; PAKAI PROJECT "ROOT"
```

```
main :
ldi r16,low(ramend)
out spl,r16                ; spl = stack pointer low
ldi r16,high(ramend)
out sph,r16



ldi zl,low(2*dat)
ldi zh,high(2*dat)  ( perintah untuk menyimpan / memnaggil data )
lpm r0,Z
inc r30 ; buat naikin register z
lpm r1,Z

mov r2,r0
mov r7,r1
add r2,r7
mov r19,r2
rcall daa
mov r2,r19
adc r3,r28

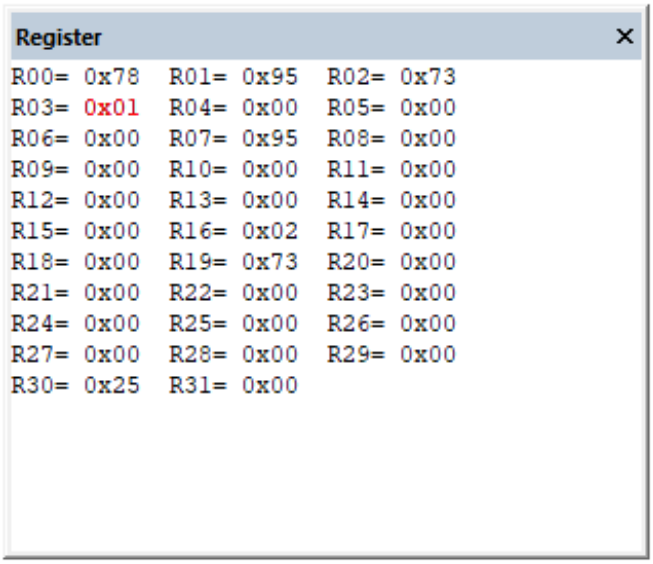
akhir: rjmp akhir

dat:
.db 0x78, 0x95

; subrutin unt koreksi hasil penjumlahan mjd. BCD.2 dgt.
; yang dilaksanakan mll. r19 sbg. acc.:
daa:    push r16;amankan isi reg. ke stack
                push r17
                push r21
                push r24
                in r16,sreg; amankan sreg
```

```
                in r17,sreg
                ldi r24,0 ; r24 hanya sbg status dr carry flag
                adc r24,r24; status carry flag sebelum koreksi
; ambil, amankan kembali sreg, uji half carry
                out sreg,r17
                brhc lup4; lompat ke lup4 jika H=0
                rcall fk_ln; satuan + 06 jika H=1
        ;       rjmp lup5
lup4:   mov r21,r19
                andi r21,$0f; ambil dgt.satuan
                cpi r21,$0a ; > 9
                brcs lup5; lompat ke lup5 jika < 9
                rcall fk_ln; satuan + 06 jika > 9
lup5:   out sreg,r17; ambil, amankan kembali sreg, uji carry
                brcc lup6; lompat ke lup6 jika C=0
                rcall fk_hn; puluhan + 60 jika C=1
        ;       rjmp lup7
lup6:   mov r21,r19
                andi r21,0xf0; ambil dgt.puluhan
                cpi r21,0xa0
                brcs lup7; lompat ke lup7 jika pul < a
                rcall fk_hn; pul + 60 jika > 9
lup7:   out sreg,r16; mengatur carry flag sblm keluar:
                clc; C = 0
                in r16,sreg ;amankan sreg stlh c=0
                cpi r24,0 ;menguji status
                breq lup8
                out sreg,r16;ambil sreg, set c
                sec; C = 1
                in r16,sreg; amankan sreg stlh c=1
lup8:   out sreg,r16;status akhir
                pop r24;kembalikan isi reg. dr stack
                pop r21
```

```
                pop r17
                pop r16
                ret
fk_ln:  ldi r21,06; koreksi dgt.satuan
                out sreg,r16
                add r19,r21
                in r16,sreg
                brcc tuh
                ldi r21,0x01
                add r17,r21
tuh:    out sreg,r16
                adc r24,r24
                ret
fk_hn:  ldi r21,$60; koreksi dgt.puluhan
                out sreg,r16
                add r19,r21
                in r16,sreg
                adc r24,r24; amankan carry pd r24
                ret ; akhir dari prog. subrutin.
```



e. Buatlah soal no 1 dengan hasil disimpan di program area

```
.include "m8535def.inc"
.org 0x00
rjmp main
        ; PAKAI PROJECT "ROOT"
main :
ldi r20,low(ramend)
out spl,r20
ldi r20,high(ramend)
out sph,r20

ldi zl,low(2*BIL)
ldi zh,high(2*BIL)
LPM R0,Z
INC R30
LPM R1,Z

mov r19,r0
add r19,r1
rcall daa
mov r2,r19
adc r3,r5



akhir: rjmp akhir

BIL :
.db 0X78, 0X95



; subrutin unt koreksi hasil penjumlahan mjd. BCD.2 dgt.
; yang dilaksanakan mll. r19 sbg. acc.:
daa:    push r16;amankan isi reg. ke stack
```

```
                push r17
                push r21
                push r24
                in r16,sreg; amankan sreg
                in r17,sreg
                ldi r24,0 ; r24 hanya sbg status dr carry flag
                adc r24,r24; status carry flag sebelum koreksi
; ambil, amankan kembali sreg, uji half carry
                out sreg,r17
                brhc lup4; lompat ke lup4 jika H=0
                rcall fk_ln; satuan + 06 jika H=1
        ;       rjmp lup5
lup4:   mov r21,r19
                andi r21,$0f; ambil dgt.satuan
                cpi r21,$0a ; > 9
                brcs lup5; lompat ke lup5 jika < 9
                rcall fk_ln; satuan + 06 jika > 9
lup5:   out sreg,r17; ambil, amankan kembali sreg, uji carry
                brcc lup6; lompat ke lup6 jika C=0
                rcall fk_hn; puluhan + 60 jika C=1
        ;       rjmp lup7
lup6:   mov r21,r19
                andi r21,0xf0; ambil dgt.puluhan
                cpi r21,0xa0
                brcs lup7; lompat ke lup7 jika pul < a
                rcall fk_hn; pul + 60 jika > 9
lup7:   out sreg,r16; mengatur carry flag sblm keluar:
                clc; C = 0
                in r16,sreg ;amankan sreg stlh c=0
                cpi r24,0 ;menguji status
                breq lup8
                out sreg,r16;ambil sreg, set c
                sec; C = 1
```

```
                in r16,sreg; amankan sreg stlh c=1
lup8:
                out sreg,r16;status akhir
                pop r24;kembalikan isi reg. dr stack
                pop r21
                pop r17
                pop r16
                ret
fk_ln:  ldi r21,06; koreksi dgt.satuan
                out sreg,r16
                add r19,r21
                in r16,sreg
                brcc tuh
                ldi r21,0x01
                add r17,r21
tuh:    out sreg,r16
                adc r24,r24
                ret
fk_hn: ldi r21,$60; koreksi dgt.puluhan
                out sreg,r16
                add r19,r21
                in r16,sreg
                adc r24,r24; amankan carry pd r24
        ret ; akhir dari prog. subrutin.
```

Register

```
R00= 0x78  R01= 0x95  R02= 0x73
R03= 0x01  R04= 0x00  R05= 0x00
R06= 0x00  R07= 0x00  R08= 0x00
R09= 0x00  R10= 0x00  R11= 0x00
R12= 0x00  R13= 0x00  R14= 0x00
R15= 0x00  R16= 0x00  R17= 0x00
R18= 0x00  R19= 0x73  R20= 0x02
R21= 0x00  R22= 0x00  R23= 0x00
R24= 0x00  R25= 0x00  R26= 0x00
R27= 0x00  R28= 0x00  R29= 0x00
R30= 0x21  R31= 0x00
```

f. Buatlah soal nomer 2 dengan hasil disimpan di program area

```
.include "m8535def.inc"
.org 0x00
rjmp main
        ; PAKAI PROJECT "ROOT"
main :
ldi r16,low(ramend)
out spl,r16             ; spl = stack pointer low
ldi r16,high(ramend)
out sph,r16

ldi zl,low(2*dat)
ldi zh,high(2*dat)
lpm r20,Z+
lpm r21,Z+
lpm r22,z+
lpm r23,z

mov r19,r20
add r19,r22
rcall daa
mov r5,r19

mov r19,r21
adc r19,r23
rcall daa
mov r6,r19

adc r7,r1

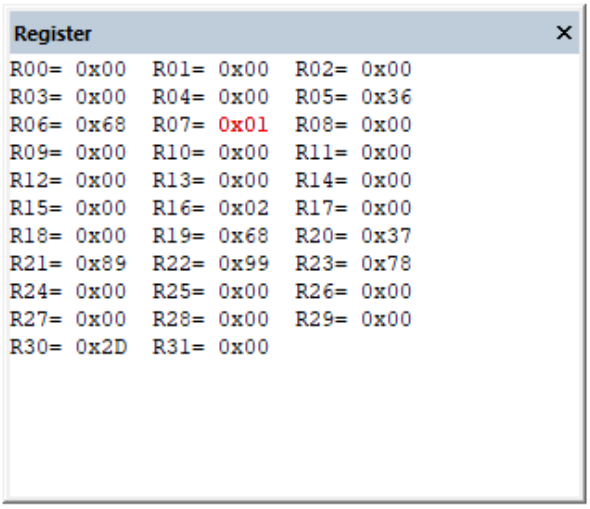
akhir: rjmp akhir

dat:
```

```
.db 0x37,0x89,0x99,0x78

; subrutin unt koreksi hasil penjumlahan mjd. BCD.2 dgt.
; yang dilaksanakan mll. r19 sbg. acc.:
daa:    push r16;amankan isi reg. ke stack
        push r17
        push r21
        push r24
        in r16,sreg; amankan sreg
        in r17,sreg
        ldi r24,0 ; r24 hanya sbg status dr carry flag
        adc r24,r24; status carry flag sebelum koreksi
; ambil, amankan kembali sreg, uji half carry
        out sreg,r17
        brhc lup4; lompat ke lup4 jika H=0
        rcall fk_ln; satuan + 06 jika H=1
    ;   rjmp lup5
lup4:   mov r21,r19
        andi r21,$0f; ambil dgt.satuan
        cpi r21,$0a ; > 9
        brcs lup5; lompat ke lup5 jika < 9
        rcall fk_ln; satuan + 06 jika > 9
lup5:   out sreg,r17; ambil, amankan kembali sreg, uji carry
        brcc lup6; lompat ke lup6 jika C=0
        rcall fk_hn; puluhan + 60 jika C=1
    ;   rjmp lup7
lup6:   mov r21,r19
        andi r21,0xf0; ambil dgt.puluhan
        cpi r21,0xa0
        brcs lup7; lompat ke lup7 jika pul < a
        rcall fk_hn; pul + 60 jika > 9
lup7:   out sreg,r16; mengatur carry flag sblm keluar:
        clc; C = 0
```

```
                in r16,sreg ;amankan sreg stlh c=0
                cpi r24,0 ;menguji status
                breq lup8
                out sreg,r16;ambil sreg, set c
                sec; C = 1
                in r16,sreg; amankan sreg stlh c=1
lup8:   out sreg,r16;status akhir
                pop r24;kembalikan isi reg. dr stack
                pop r21
                pop r17
                pop r16
                ret
fk_ln:  ldi r21,06; koreksi dgt.satuan
                out sreg,r16
                add r19,r21
                in r16,sreg
                brcc tuh
                ldi r21,0x01
                add r17,r21
tuh:    out sreg,r16
                adc r24,r24
                ret
fk_hn:  ldi r21,$60; koreksi dgt.puluhan
                out sreg,r16
                add r19,r21
                in r16,sreg
                adc r24,r24; amankan carry pd r24
                ret ; akhir dari prog. subrutin.
```



A. LAB 2_1

```
.include "m8535def.inc"
.org 0x00
rjmp main
        ; PAKAI PROJECT "ROOT"
main :
ldi r16,low(ramend)
out spl,r16
ldi r17,high(ramend)
out sph,r17

ldi r20,0XFF
mov r25,r20
andi R25,0XF0 ; AMBIL JADI 16 AN, PAKE LSR
lsr R25
lsr R25
lsr R25
lsr R25
mov r19,r25
add r19,r30;karena flagnya udh terpengaruh sama r19 (carry)
rcall daa
```

```
mov r21,r19
ldi r16,4
lup:mov r19,r21
        add r19,r19
        rcall daa
        mov r21,r19
        mov r19,r22
        adc r19,r19
        rcall daa
        mov r22,r19
        dec r16
        BRNE lup
mov r25,r20
andi r25,0X0f
mov r19,r25
add r19,r30
rcall daa
mov r25,r19
mov r19,r21
add r19,r25
rcall daa
mov r21,r19
mov r19,r22
adc r19,r30
rcall daa
mov r22,r19

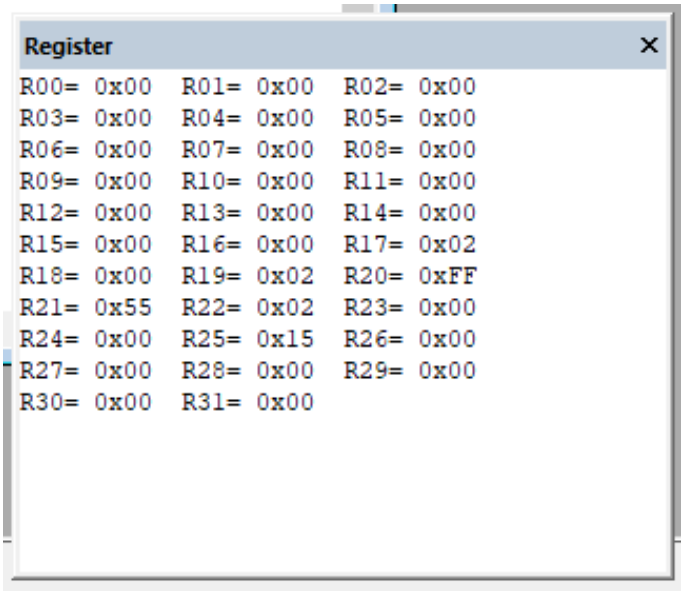


akhir: rjmp akhir

; subrutin unt koreksi hasil penjumlahan mjd. BCD.2 dgt.
; yang dilaksanakan mll. r19 sbg. acc.:
daa:    push r16;amankan isi reg. ke stack
```

```
                push r17
                push r21
                push r24
                in r16,sreg; amankan sreg
                in r17,sreg
                ldi r24,0 ; r24 hanya sbg status dr carry flag
                adc r24,r24; status carry flag sebelum koreksi
; ambil, amankan kembali sreg, uji half carry
                out sreg,r17
                brhc lup4; lompat ke lup4 jika H=0
                rcall fk_ln; satuan + 06 jika H=1
        ;       rjmp lup5
lup4:   mov r21,r19
                andi r21,$0f; ambil dgt.satuan
                cpi r21,$0a ; > 9
                brcs lup5; lompat ke lup5 jika < 9
                rcall fk_ln; satuan + 06 jika > 9
lup5:   out sreg,r17; ambil, amankan kembali sreg, uji carry
                brcc lup6; lompat ke lup6 jika C=0
                rcall fk_hn; puluhan + 60 jika C=1
        ;       rjmp lup7
lup6:   mov r21,r19
                andi r21,0xf0; ambil dgt.puluhan
                cpi r21,0xa0
                brcs lup7; lompat ke lup7 jika pul < a
                rcall fk_hn; pul + 60 jika > 9
lup7:   out sreg,r16; mengatur carry flag sblm keluar:
                clc; C = 0
                in r16,sreg ;amankan sreg stlh c=0
                cpi r24,0 ;menguji status
                breq lup8
                out sreg,r16;ambil sreg, set c
                sec; C = 1
```

```
                in r16,sreg; amankan sreg stlh c=1
lup8:
                out sreg,r16;status akhir
                pop r24;kembalikan isi reg. dr stack
                pop r21
                pop r17
                pop r16
                ret
fk_ln:  ldi r21,06; koreksi dgt.satuan
                out sreg,r16
                add r19,r21
                in r16,sreg
                brcc tuh
                ldi r21,0x01
                add r17,r21
tuh:    out sreg,r16
                adc r24,r24
                ret
fk_hn:  ldi r21,$60; koreksi dgt.puluhan
                out sreg,r16
                add r19,r21
                in r16,sreg
                adc r24,r24; amankan carry pd r24
                ret ; akhir dari prog. subrutin.
```



B. LAB 2_2

```
.include "m8535def.inc"
.org 0x00
rjmp main
        ; PAKAI PROJECT "ROOT"
main :
ldi r20,low(ramend)
out spl,r20
ldi r20,high(ramend)
out sph,r20

ldi r16,0x77  ;register input data
mov r20,r16
andi r20,0xf0 ; mengambil nilai puluhan
lsr r20
lsr r20
lsr r20
lsr r20
ldi r21,0x0a
mul r20,r21
```

```
mov r17,r0


mov r20,r16
andi r20,0x0f


add r17,r20
akhir: rjmp akhir
```

